

Behavior monitoring for an overinformative support agent

A deployed monitor that samples traces, runs a frozen judge, corrects prevalence for judge error, and preserves auditable evidence.

FROZEN JUDGE

TRACE-LEVEL

CI AUTOMATED

Prepared September 28, 2026

Repository commit [fa2b6c8](#) | Workflow run [36467153998](#)

What we built and what it tells us

A production-shaped monitor with explicit sampling, calibration, uncertainty, and CI evidence.

24-HOUR RUN

91 traces

88 conversations scanned

RANDOM SAMPLE

18 traces

14 flagged; raw rate 77.8%

RISK SAMPLE

14 traces

8 flagged; raw rate 57.1%

CORRECTED ESTIMATE

88.4%

95% CI: 33.3% to 100%

INTERPRETATION

The failure mode is common enough to act on.

- The corrected point estimate is well above the 15% action threshold.
- The interval remains wide because the frozen judge has limited specificity and the samples are small.
- Treat the direction as strong operational evidence, while avoiding claims about the exact population rate.

RECOMMENDATION

Ship the response constraint

Require concise order-status answers, disclose only the requested facts, and offer support escalation when policy is unavailable.

Then monitor again

Increase the random sample and re-calibrate the judge so the confidence interval can support a sharper release decision.

Before and after: prevalence moved up

The same frozen judge and calibration were used for both runs; sample composition changed slightly.

MEASURE	BEFORE	AFTER	CHANGE
Random sample flagged	7 / 10 (70.0%)	9 / 10 (90.0%)	+20.0 pp
Policy-risk sample flagged	5 / 8 (62.5%)	7 / 10 (70.0%)	+7.5 pp
Corrected prevalence	72.2% [0%, 100%]	100% [59.3%, 100%]	+27.8 pp



HOW TO READ THIS

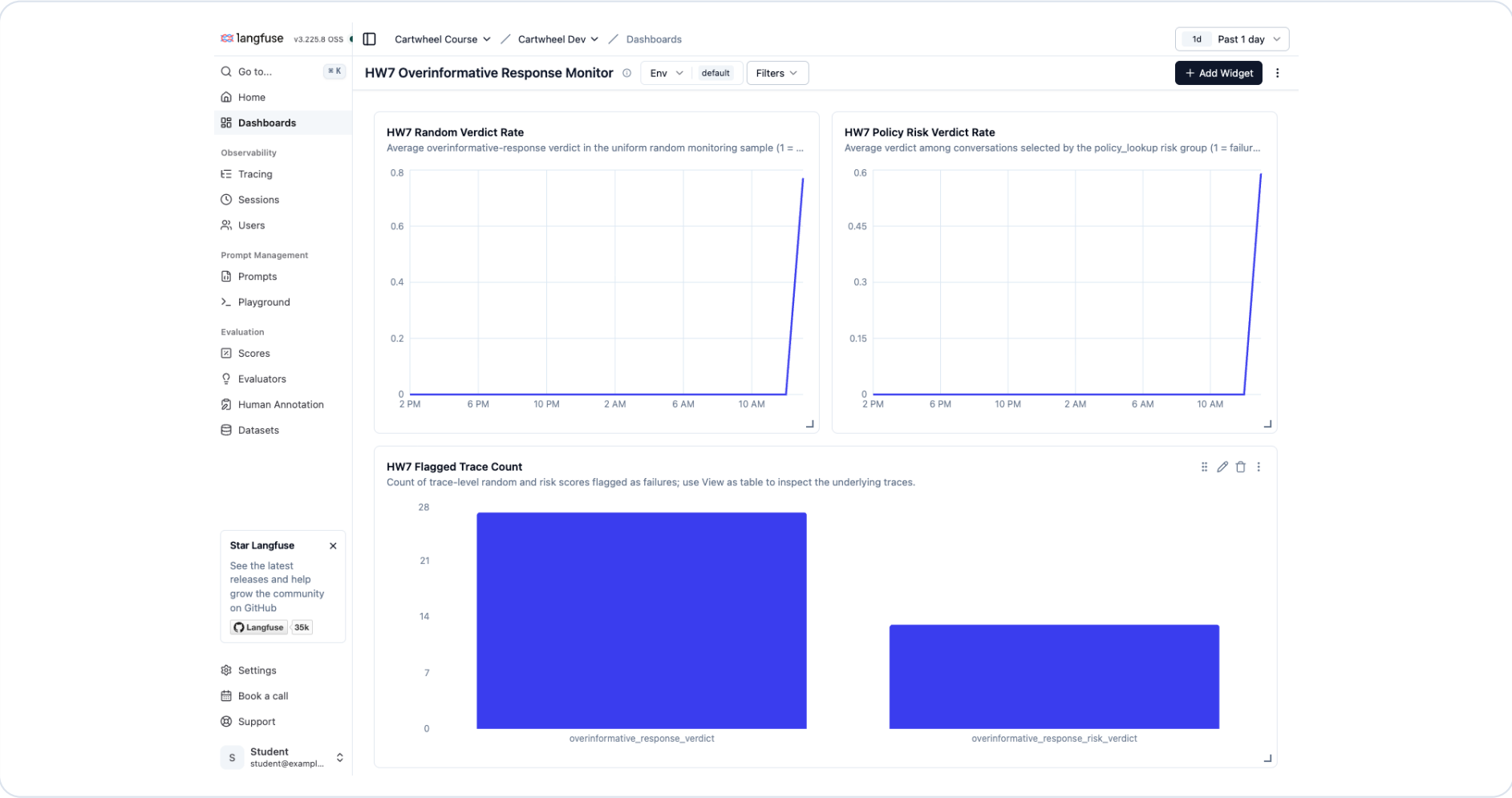
Both point estimates exceed the 15% threshold, but their confidence intervals overlap. The observed increase is therefore a signal to investigate, not proof that the population rate worsened.

Frozen evaluator

overinformative_response-v0 | gpt-4o-mini
Sensitivity 83.3% | specificity 64.7%

Live Langfuse monitoring dashboard

Three widgets make the failure rate and flagged traces inspectable in the same observability surface.

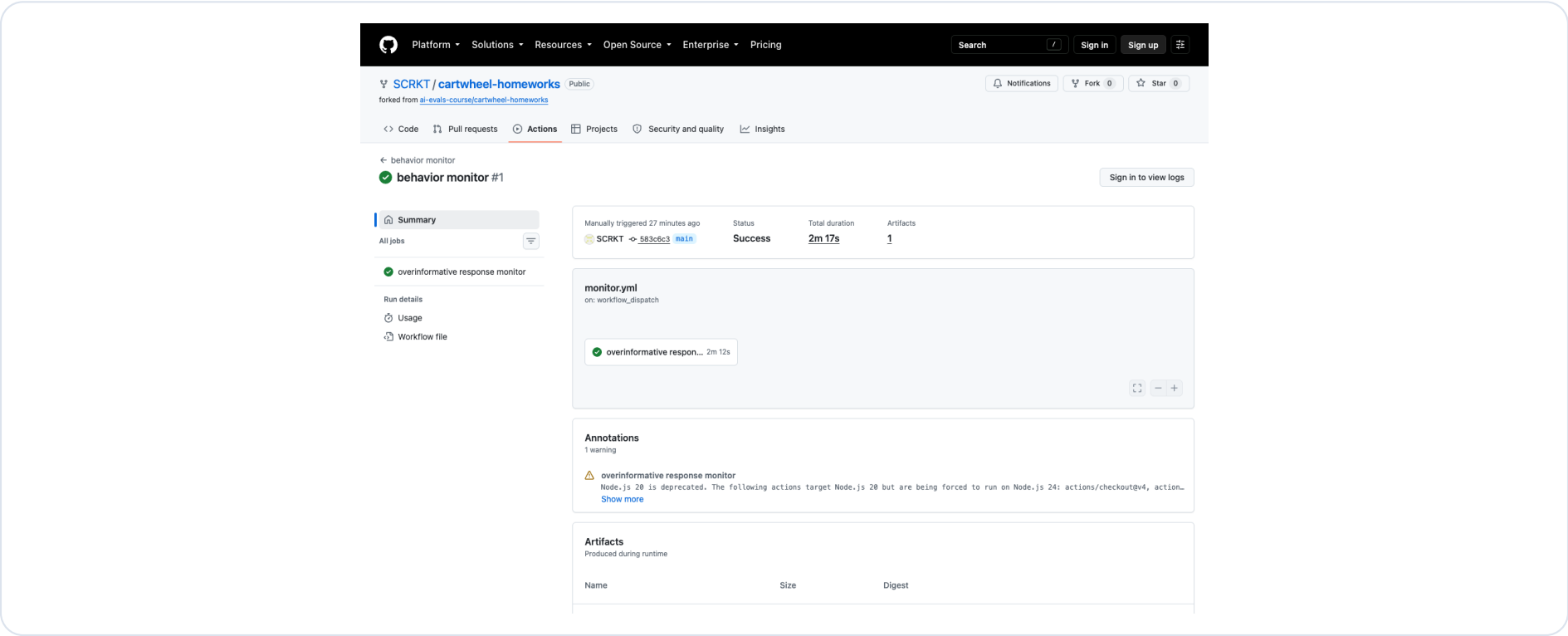


SCOPE NOTE

The screenshot shows dashboard aggregates over the selected day. Experiment summaries use unique trace IDs from saved artifacts, so repeated score writes do not inflate the reported samples.

Automated monitor completed successfully

The self-hosted runner executed the frozen monitor and retained a compact evidence artifact.



WORKFLOW

Success

2m 17s total duration

ARTIFACT

1 retained

behavior-monitor-36467153998

SCORES WRITTEN

33

30 unique judge calls

IDEMPOTENCY

Verified

unique random, risk, prevalence IDs

Why the result is reproducible

The monitor records the data path, evaluator version, correction method, and operational evidence.

1 Select

Uniform random traces estimate prevalence; policy_lookup traces form a targeted risk slice.

2 Judge

A frozen prompt and gpt-4o-mini produce trace-level binary verdicts with stable score names.

3 Correct

The random-sample rate is adjusted using calibrated sensitivity and specificity, then clipped to $[0, 1]$.

4 Quantify

A bootstrap interval exposes uncertainty rather than treating the corrected point estimate as exact.

5 Persist

JSON, JSONL, SVG, and Langfuse scores preserve both the latest result and history.

6 Automate

GitHub Actions runs the same command on a self-hosted runner and uploads the evidence artifact.

NEXT ITERATION

Increase the random sample, label fresh examples, and re-calibrate the judge. Keep the 15% threshold, but make release decisions from the confidence interval and reviewed examples together.